

## Contact

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www.linkedin.com/in/alan-palayil  
(LinkedIn)  
alanp13.github.io/web-portfolio/  
(Personal)

## Top Skills

Vulnerability Management  
Graph Data Algorithms  
GDS Library

## Languages

English (Native or Bilingual)  
Hindi (Native or Bilingual)  
Malayalam (Limited Working)  
Marathi (Full Professional)

## Certifications

Beginning C++ Programming - From  
Beginner to Beyond  
The Bits and Bytes of Computer  
Networking  
Neo4j Graph Data Science  
Certification  
The Complete Cyber Security  
Course : All Modules  
Meta Front-End Developer  
Professional Certificate

## Honors-Awards

Dean's List Recipient Fall 2020  
Dean's List Recipient Spring 2023  
Dean's List Recipient Spring 2022  
Dean's List Recipient Fall 2021  
Dean's List Recipient Spring 2021

# Alan Biju Palayil

Data Engineer @ Genworth | PhD Student | M.S. Cybersecurity  
Engineering | FinTech AI Systems, Data Science, & Cybersecurity  
Richmond, Virginia, United States

## Summary

I currently hold the position of Application Development Analyst in Investments IT at Genworth, where I oversee accounting support and spearhead technology initiatives across multiple accounting asset classes, including Fixed Income (FI), CML, and Derivatives Processing, as well as Cloud Migration and Transformation.

Since joining Genworth in 2023, I have taken a proactive role in streamlining legacy systems and facilitating large-scale transitions to the Azure Cloud environment. Over the past year, I have been instrumental in rehosting and modernizing legacy platforms in Azure, enhancing process functionality, and strengthening Investments IT's cloud-based accounting capabilities through the implementation of robust ETL pipelines.

I serve as the IT Owner for the CML project, acting as the primary technical authority responsible for ensuring system stability, driving enhancements, and enabling cross-team collaboration. My work operates at the intersection of finance, cloud infrastructure, and application development, leveraging hands-on expertise in automation, data pipelines, ETL workflows, and investment accounting platforms. I collaborate closely with Accounting, Investment Operations, and Controllershship to translate business requirements into resilient and scalable technical solutions.

I hold a Master's degree in Cybersecurity Engineering and a Bachelor's degree in Computer and Cybersecurity Engineering from the Illinois Institute of Technology, where I consistently earned Dean's List recognition. I am currently pursuing a Doctorate in Information Technology, with a focus on secure systems, cloud architectures, and data-driven financial platforms.

I am committed to building systems that are resilient, secure, and future-ready, and I thrive on solving complex challenges where technology delivers measurable business impact.

Open to conversations around cloud migration, financial systems modernization, and large-scale application development.

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## Experience

### Genworth

3 years 2 months

#### Data Engineer

July 2026 - Present (2 months)

Richmond, VA

Lead data engineering and application modernization for enterprise investment accounting systems supporting Commercial Mortgage Loans and Derivatives.

#### Level P2

#### Core Responsibilities & Contributions

- Design and maintain Python- and SQL-based data pipelines, ETL workflows, interfaces, extracts, reconciliations, and JSON message-generation services.
- Develop cloud-based processing solutions using Microsoft Azure, Azure Synapse Analytics, Azure Data Lake Storage, and notebooks.
- Serve as the IT owner for Commercial Mortgage Loan workflows within PAM, overseeing enhancements, production support, data validation, and operational stability.
- Build automated file-transfer, monitoring, retry, notification, dependency-management, and reconciliation processes to reduce manual intervention.
- Perform solution design, development, testing, deployment, incident investigation, root-cause analysis, defect resolution, and performance optimization.
- Partner with Investments Operations, Accounting, Controllershship, Infrastructure, QA, vendors, and engineering teams to deliver reliable technical solutions.
- Follow Agile, SDLC, version-control, documentation, and enterprise change-management practices.

#### Technical Proficiencies

- Languages & Data Formats: Python, SQL, JSON, XML
- Cloud & Data Platforms: Microsoft Azure, Azure Synapse Analytics, Azure Data Lake Storage, PostgreSQL
- Integration & Processing: Markit EDM, Informatica, ETL/ELT workflows, REST APIs, secure file transfer, enterprise batch processing
- Engineering Tools: Git, Azure DevOps, AppWorx, notebooks
- Methodologies: Agile, SDLC, data validation, reconciliation, root-cause analysis, production support, performance optimization, technical documentation

### Application Development Analyst

December 2025 - July 2026 (8 months)

Richmond, VA

Led application engineering and modernization initiatives for enterprise investment accounting systems, owning Commercial Mortgage Loan workflows within PAM and supporting the migration of legacy Derivatives processing solutions to Microsoft Azure.

Level: P2

### Core Responsibilities & Contributions

- Served as the IT owner for CML processing within PAM, providing technical oversight, enhancement planning, production support, and issue resolution.
- Supported the migration and modernization of legacy message-generation and investment-processing applications in Microsoft Azure.
- Developed Python applications, SQL queries, ETL workflows, data interfaces, extracts, reconciliation processes, and JSON message-generation services.
- Supported CML migration workflows involving loan, collateral, fee, payment, interface, and reconciliation data.
- Built automated file-transfer, monitoring, retry, notification, validation, and dependency-management processes.
- Investigated production incidents, performed root-cause analysis, and implemented sustainable fixes that improved reliability and auditability.
- Collaborated with Investments Operations, Accounting, Controllershship, Infrastructure, QA, vendors, and project stakeholders.
- Participated in Agile planning, technical design, testing, deployment, documentation, and production releases.

### Technical Proficiencies

- Languages & Tools: Python, SQL, Git, AppWorx

- Databases & Platforms: PostgreSQL, Azure Synapse Analytics, Azure Cloud Services
- Data Integration: Markit EDM, Informatica, enterprise ETL workflows
- Problem Solving: Application troubleshooting, cloud performance tuning, data validation, and process optimization

### Associate Application Development Analyst

July 2023 - December 2025 (2 years 6 months)

Richmond, VA

Developed and supported automation, data interfaces, and reporting solutions for enterprise investment accounting workflows, replacing manual processes with repeatable and maintainable Python- and SQL-based solutions.

Level: P1

### Core Responsibilities & Contributions

- Created Python scripts and SQL queries for data transformation, validation, reconciliation, reporting, file processing, and recurring operational tasks.
- Maintained and supported applications used by Investments Trading, Accounting, and Fixed-Income operations.
- Developed automated reports, extracts, system interfaces, ETL workflows, and enterprise batch-processing solutions.
- Monitored scheduled processing, investigated application and data issues, and assisted with production incident resolution.
- Identified opportunities to replace manual processes with controlled, repeatable, and maintainable automation.
- Performed unit testing, system testing, data validation, defect investigation, and technical documentation.
- Collaborated with Investments Operations, Accounting, Controllershship, QA, and engineering teams to gather requirements and deliver enhancements.
- Participated in Agile sprint planning, backlog refinement, development, testing, and release activities.

### Technical Proficiencies

- Languages: Python, SQL
- Databases & Platforms: PostgreSQL, Microsoft Azure, Azure Synapse Analytics
- Data Integration: Markit EDM, Informatica, ETL workflows, enterprise batch processing
- Engineering Tools: Git, AppWorx, Azure DevOps

- Methodologies: Agile, SDLC, application troubleshooting, data validation, testing, root-cause analysis, technical documentation, process automation

## Illinois Institute of Technology

### Student Assistant

August 2021 - May 2023 (1 year 10 months)

Chicago, IL

Provided frontline customer service, administrative assistance, access-control support, and incident coordination for university residence halls while helping maintain a safe and responsive residential environment.

### Core Responsibilities & Contributions

- Managed residence-hall front-desk operations and assisted students, staff members, residents, and visitors with questions, requests, and facility-related concerns.
- Verified resident and visitor information, maintained access and visitor records, issued and collected keys or authorized equipment, and followed university security procedures.
- Received and processed resident packages, maintained accurate records, and assisted with package distribution.
- Communicated resident concerns, maintenance issues, emergencies, and policy matters to resident advisors, professional residence-life staff, campus safety personnel, and other authorized departments.
- Documented incidents and operational activities while maintaining confidentiality and complying with university housing policies and safety requirements.
- Supported evening and overnight residence-hall operations, responding professionally to time-sensitive requests and escalating emergencies when necessary.
- Used university housing systems, Microsoft Office applications, email, telephones, scanners, printers, and other standard office equipment.

## ITHENA

### Full Stack Developer Intern

June 2020 - August 2020 (3 months)

Richmond, VA

Developed full-stack and IoT software components for a connected street-lighting prototype, including device communication, server integration, and a web-based monitoring and control interface.

### Core Responsibilities & Contributions

- Worked under the supervision of a senior software engineer within the Product Development department to design and develop an IoT street-lamp prototype.
- Developed C# application components and implemented MQTT-based messaging for communication between connected devices and server-side systems.
- Configured TCP/IP communication to support direct device-to-server data transfer without relying on a restricted third-party automation platform.
- Designed and developed a browser-based monitoring and control interface using JavaScript, HTML, and CSS.
- Built services and application components that communicated with server-side technologies and supported real-time device data exchange.
- Participated in software design, coding, debugging, testing, deployment preparation, and research into networking and IoT communication techniques.
- Created technical guidelines, test reports, user documentation, and supporting materials for the prototype.
- Collaborated with engineering team members to troubleshoot software and network issues and improve system functionality.
- Developed internal tools and processes supporting quality assurance, deployment, testing, and technical support activities.

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## Education

### University of the Cumberlands

Doctor of Philosophy - PhD, Information Technology · (August 2025 - May 2029)

### Illinois Institute of Technology

Master of Science - MS, Cybersecurity Engineering · (May 2022 - August 2023)

### Illinois Institute of Technology

Bachelor of Science - BS, Computer and Cyber-security Engineering · (August 2019 - August 2023)

### Fr. Agnel Multipurpose School & Jr. College

High School Diploma, Computer Science · (June 2007 - March 2019)